



Scientific Notation

Day 5 of 20 · Theme: Itasca State Park

MINNESOTA STANDARDS (MDE 2022)

Standard: 9.3.5.1

LEARNING OBJECTIVES & SUCCESS CRITERIA

- I can apply the lesson's core skill with fluency and check my answers for reasonableness.
- I can explain my reasoning using math vocabulary.
- I can connect the math to the lesson's real-world career and cultural context.

ENGAGEMENT & PACING

Strategy: Think-Write-Pair-Share **Practice Format:** Merit Time Game — Power of 10 Lane

Duration: 50 minutes

TEACHER ANNOTATIONS — PRACTICE ANSWERS (VERIFIED)

Scaffold (Tier 1) Prompt: Write 3.2×10^4 in standard form. *Model answers:*

• 32,000 **On-Level (Tier 2) Prompt:** Write 0.00052 using scientific notation. Multiply 2×10^3 by 3×10^4 . *Model answers:*

• 5.2×10^{-4}

• 6×10^7 **Extension (Tier 3) Prompt:** A cell phone has 6.4×10^{10} bytes of storage. Write it in standard form. *Model answers:*

• 64,000,000,000

EXIT TICKET — ANSWER & SCORING

Exit ticket: Show your work for full credit *Scoring: 1 point per exit ticket; mastery = 4 of 4. Collect before the final 5 minutes; address misconceptions next lesson's bell ringer.*

DIFFERENTIATION (3 TIERS)

- Tier 1 (Support): Use tiered practice 'go'; provide a GEMDAS keyword bank and sentence starters.
- Tier 2 (On-Level): Work 'set' problems; partner check before whole-class reveal.
- Tier 3 (Challenge): Complete 'extra' extension; ask students to create their own verbal phrase.

CULTURE & CAREER CONNECTION

- Culture: Ethiopian (Ethiopia) Ethiopian Americans are a growing Minnesota community, centered in Cedar-Riverside. Ethiopian academic tradition dates to antiquity, including the Ge'ez numeral system used for counting and record-keeping.

- Career: Best Buy — Richfield | Inventory Specialist | HS | median \$37,330 *O*NET SOC: 53-7065.00*

MERIT EDUCATION GROUP™ | Intermediate Algebra | Unit 1 | Lesson 5